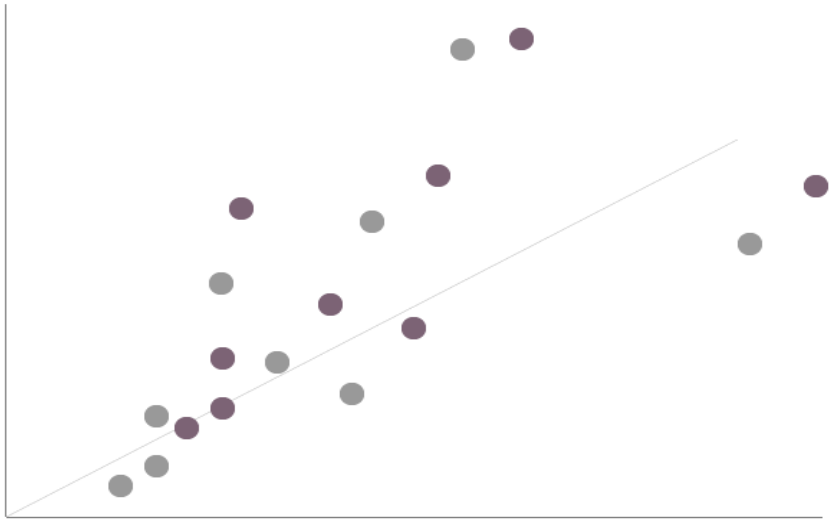


Research Designs

Yashpal Jogdand

AHUL261

What do you think?

1. **The more highly educated a woman is the fewer children she is likely to have.**

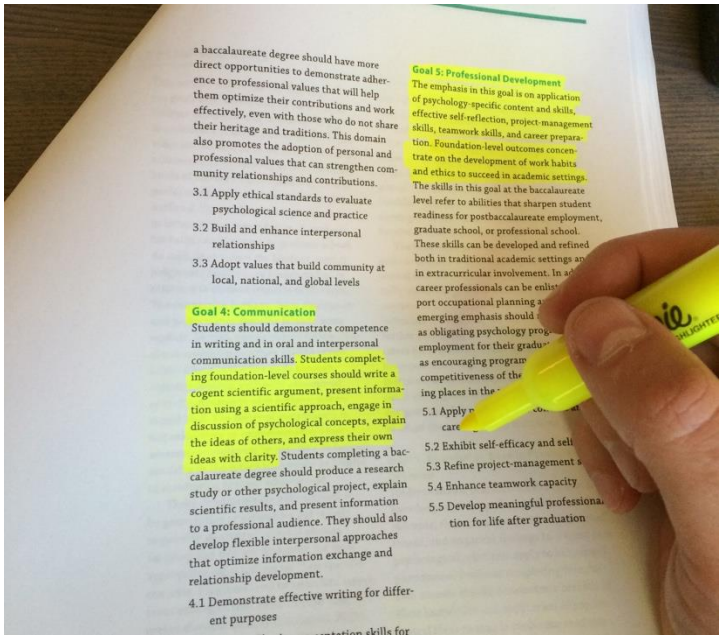
2. **The less time students spend on Facebook the better their grades are.**

Overview

- **Experimental Research**
- Correlational Designs
- Surveys
- Quasi-Experimental Design
- Longitudinal Designs
- Trade-offs
- Methods

Experimental Research

- Does highlighting improve retention of information?
- Randomly draw names
- Assign them to highlight
- Other half no highlighting

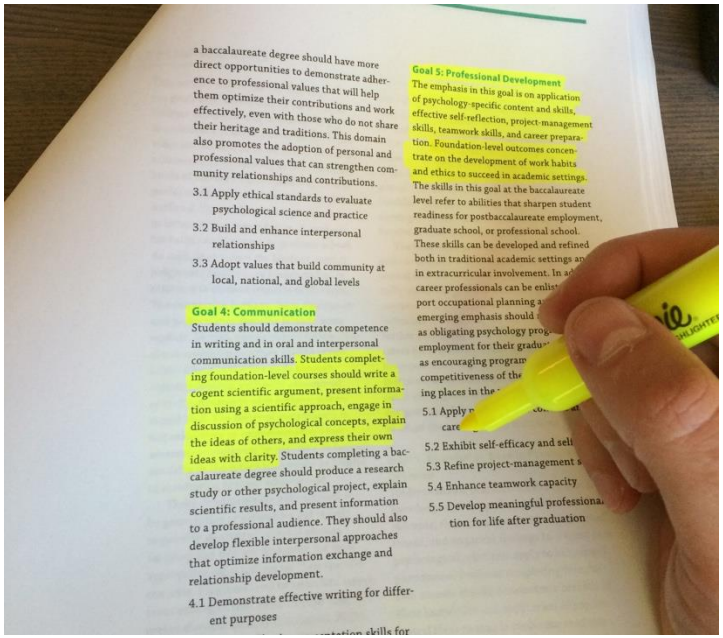


Experimental Research

- Does highlighting improve retention of information?

Discussion

- Research or intuition?
- Independent variable?
- Dependent variable?
- Importance of random assignment?



Experimental Research

Placebo Effect



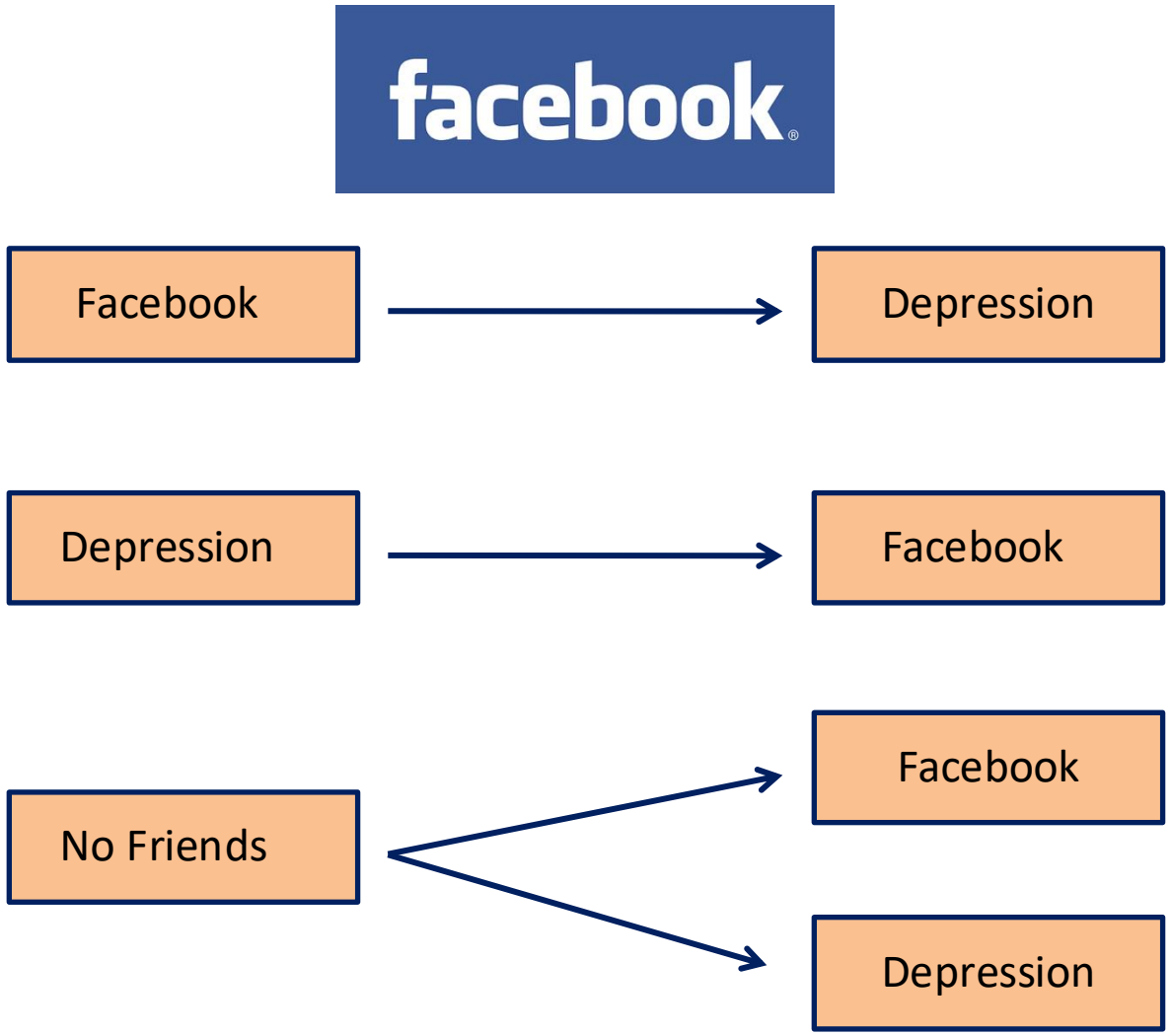
Discussion

- Confounding variables?
- Participant demands affect outcomes?
- Experimenter expectations influence outcomes?

Overview

- Experimental Research
- **Correlational Designs**
- Surveys
- Quasi-Experimental Design
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Correlation is not Causation



Activity: Correlation Coefficient

Relationships

- Time on social media & exam scores
- Animal-lover & owning a cat
- Having a sweet tooth & number of desserts consumed

Correlation Coefficient

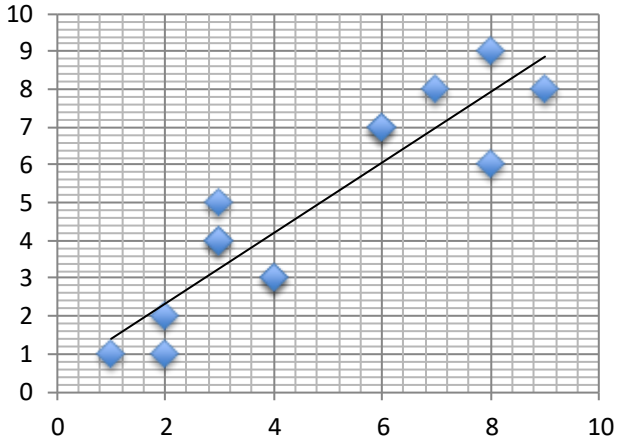
The correlation coefficient measures the strength and the direction of a linear relationship between two variables and is designated with the lower-case letter "r".

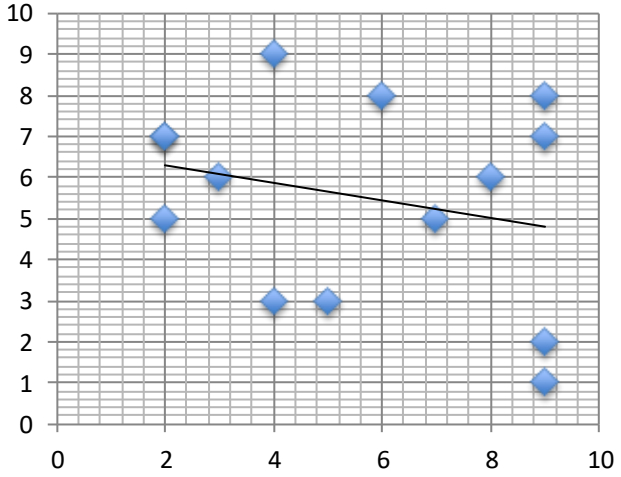
r=.75, r= -.50, r= .20

Correlational Designs

Discussion

- Which correlation is stronger? Why?
- Which correlation is positive? Why?
- Which graph represents
  - r= -.21? r=.90?
  - Highlighting study?
  - Caffeine study?
- Why can't a researcher say beliefs about caffeine cause greater alertness?





Overview

- Experimental Research
- Correlational Designs
- **Surveys**
- Quasi-Experimental Design
- Longitudinal Designs
- Trade-offs
- Methods

Surveys

Survey:

- A way of systematic observation
- Asking large numbers of people to respond to questions about their attitudes or behavior.

e.g. *student satisfaction survey*

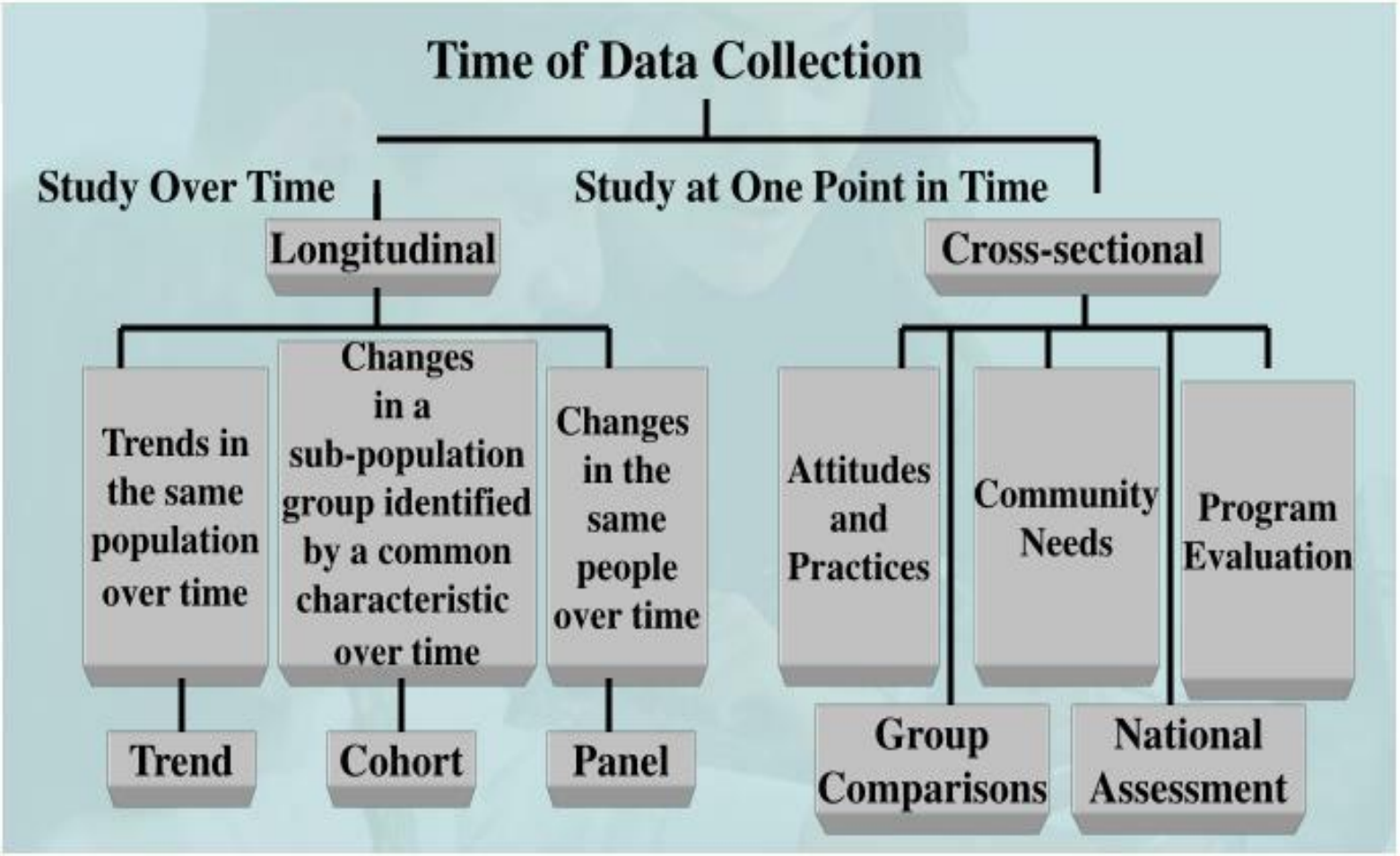
Important Issues -

Sampling (the people who participate must be representative of the larger population)

Wording (The way in which the items are worded can exert strong effects on the outcomes obtained)

- Two types- *Cross-sectional and Longitudinal*

What are the types of survey designs?

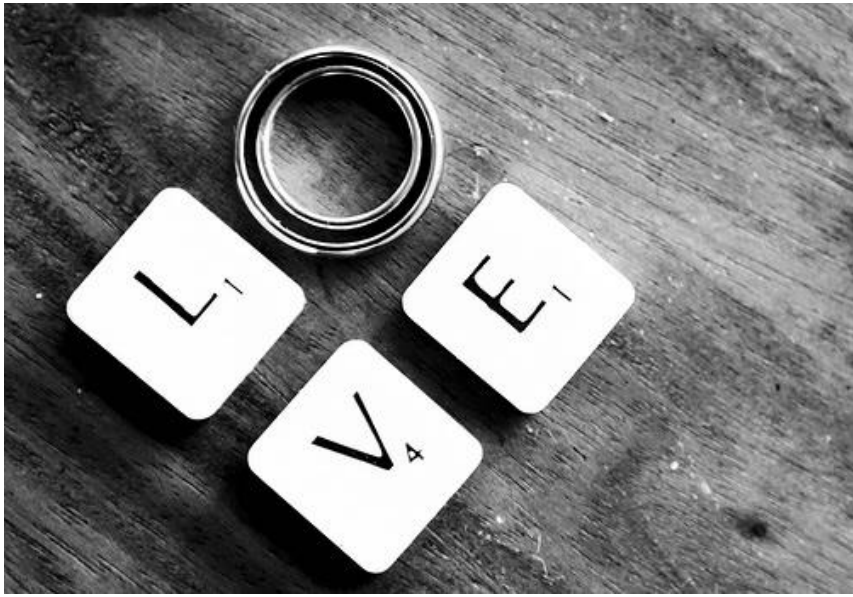


Overview

- Experimental Research
- Correlational Designs
- Surveys
- **Quasi-Experimental Design**
- Longitudinal Designs
- Trade-offs
- Methods

Quasi-Experimental Design

- No random assignment
- Different independent variables – fixed i.e. can not be manipualated/changed
- Problems?



Overview

- Experimental Research
- Correlational Designs
- Surveys
- Quasi-Experimental Design
- **Longitudinal Designs**
- Trade-offs
- Methods

Longitudinal Studies

- Involve repeated observations of the same variables over a long period of time.

- Captures changes over time, strengthens confidence in findings

- But are very expensive to conduct, attrition rates



Overview

- Experimental Research
- Correlational Designs
- Surveys
- Quasi-Experimental Design
- Longitudinal Designs
- **Methods & Trade-offs**

Methods & Trade-offs

- Which of the previously discussed research designs is the best?

- Trade-offs - no such thing as the “perfect” research design.

Reading News items Critically

- News report says - “Diet soda leads to weight gain.”
- Media’ tendency to sensationalize things
- **How did the researcher discover this?**
- **Show me the data!**



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